

Wenqing Hu

CONTACT INFORMATION	112 Rolla Building, Department of Mathematics and Statistics, Missouri University of Science and Technology, (formerly University of Missouri, Rolla) 400W, 12th St., Rolla, MO, 65409-0020, USA.	<i>Mobile:</i> (573) 612-5268 <i>Office:</i> (573) 341-4650 <i>E-mail:</i> huwen@mst.edu <i>Website:</i> https://huwenqing0606.github.io/
ACADEMIC EMPLOYMENT	Missouri University of Science and Technology. (formerly University of Missouri, Rolla) Associate Professor (with tenure), Assistant Professor (tenure-track), University of Minnesota, Twin Cities. Postdoctoral Associate, Mentor: Professor Vladimír Šverák.	Rolla, Missouri, USA. 06/2022 – present. 07/2016 – 05/2022. Minneapolis, Minnesota, USA. 08/2013 – 06/2016.
EDUCATION	University of Maryland, College Park. Ph.D. in Mathematics, Dissertation Topic: “Asymptotic Problems in Stochastic Processes and Differential Equations”. Advisor: Professor Mark Freidlin. Peking University. B.S. in Mathematics, Thesis Topic: “From Markov processes to martingales and independent increment processes”. Advisor: Professor Yong Liu.	College Park, Maryland, USA. 08/2008 – 05/2013. Beijing, P.R.China. 09/2004 – 07/2008.
RESEARCH INTERESTS	My mathematical research focuses on the <i>asymptotic and long-time behavior of complex stochastic systems</i> . I am also interested in developing <i>real-world engineering systems that involve probabilistic modeling</i> .	
PREPRINTS IN SUBMISSION / REVISION	• Wenqing Hu, Coadjoint averaging and cross-scale fluxes in a fast-slow stochastic Euler-Arnold system on $SU(N)$. https://arxiv.org/abs/2607.17943 • Jiali Zhang, Weiwei Cui, Wenqing Hu, An Optimization Framework for Manufacturing System with Scheduling. Submitted. • Jiali Zhang, Long, Y., Liu, J., Das, R., Zhang, H., Wenqing Hu, Donald Wunsch, RAM-DQN: Joint Optimization of UAV Route and Dwell Time under Uncertainty. In Revision. • Dihang Guan, Hui He, Wenqing Hu, Jiaojiao Yang, Wave propagation for 1-dimensional reaction-diffusion equation with nonzero random drift. Preprint. https://arxiv.org/abs/2512.21906 • Haotian Chen, Wenqing Hu, Siyuan Wang, Rui Bo, Deep Reinforcement Learning Based Arbitrage Strategy Development for Local Energy Storage System Considering Look-Ahead Commitment:	

Novel State Design and Robustness Analysis. Submitted.

- Wenqing Hu, Hong Qian, On the Posterior Distribution of a Random Process Conditioned on Observing the Empirical Frequencies: the i.i.d and finite Markov chain case. Preprint. Submitted.

arXiv:2202.11780 [math.PR]

JOURNAL
PUBLICATIONS

- Tianyi Liu, Zhenfei Zhang, Yuncong Zhang, Wenqing Hu, Ye Zhang, *Ceno*: Non-uniform, Segment and Parallel Zero-Knowledge Virtual Machine. *Journal of Cryptology*, **38**, 17 (2025).

DOI: <https://doi.org/10.1007/s00145-024-09533-2>

- Haoyu Wang, Xiaoliang Gan, Wenqing Hu, Ping Ao, The generalized Lyapunov function as Ao's potential function: Existence in dimensions 1 and 2, *Journal of Applied Analysis and Computation*, Volume **13**, Number 1, February 2023, pp. 359-375.

DOI: 10.11948/20220149

- Haoyu Wang, Xiaoliang Gan, Wenqing Hu, Ping Ao, Fundamental Structure of General Stochastic Dynamical Systems: High-Dimension Case, *Journal of Mathematics*, Volume 2022 Article ID 2596074.

DOI: <https://doi.org/10.1155/2022/2596074>

- Jiaojiao Yang, Zeyi Sun, Wenqing Hu, Louis Steinmeister, Joint Control of Manufacturing and Onsite Microgrid System via Novel Neural-Network Integrated Reinforcement Learning Algorithms. Volume **315**, 1 June 2022, 118982.

DOI: <https://doi.org/10.1016/j.apenergy.2022.118982>

- Wai-Tong (Louis) Fan, Wenqing Hu, Grigory Terlov, Wave propagation for reaction-diffusion equations on infinite random trees. *Communications in Mathematical Physics*, **384**, Issue 1, April 2021, pages 109-163.

DOI: 10.1007/s00220-021-04085-z. arXiv:1907.12962[math.PR]

- Md Monirul Islam, Zeyi Sun, Ruwei Qin, Wenqing Hu, Haoyi Xiong, Kaibo Xu, Flexible energy load identification in intelligent manufacturing for demand response using a neural network integrated particle swarm optimization. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, online first, 2020.

DOI: 10.1177/0954406220933652.

- Jiaojiao Yang, Wenqing Hu, Chris Junchi Li, On the fast convergence of random perturbations of the gradient flow. *Asymptotic Analysis*, Volume 122, 2021, pages 371-393.

DOI 10.3233/ASY-201622. arXiv:1706.00837[math.PR].

- Md Monirul Islam, Xiao Zhong, Zeyi Sun, Haoyi Xiong, Wenqing Hu, Real-Time Frequency Regulation Using Aggregated Electric Vehicles in Smart Grid, *Computers & Industrial Engineering*, Volume 134, August 2019, pages 11-26.

DOI: 10.1016/j.cie.2019.05.025.

- Wenqing Hu, On the long time behavior of a perturbed conservative system with degeneracy. *Journal of Theoretical Probability*, Volume 33, pp.1266-1295, 2020. (Published online on 11, May 2019.).

DOI: 10.1007/s10959-019-00911-2. [arXiv:1808.01510\[math.PR\]](#).

- Wenqing Hu, Michael Salins, Konstantinos Spiliopoulos, Large deviations and averaging for systems of slow-fast stochastic reaction-diffusion equations. *Stochastics and Partial Differential Equations: Analysis and Computations*, December 2019, Volume 7, Issue 4, pp. 808-874.

DOI: 10.1007/s40072-019-00140-y. [arXiv:1710.02618\[math.PR\]](#).

- Wenqing Hu, Chris Junchi Li, Lei Li, Jian-Guo Liu, On the diffusion approximation of nonconvex stochastic gradient descent. *Annals of Mathematical Science and Applications*, Vol. 4, No. 1 (2019), pp. 3-32.

DOI: <http://dx.doi.org/10.4310/AMSA.2019.v4.n1.a1>. [arXiv:1705.07562v2\[stat.ML\]](#).

- Wenqing Hu, Chris Junchi Li, A convergence analysis of perturbed compositional gradient flow: averaging principle and normal deviations. *Discrete and Continuous Dynamical Systems, Series A*, **38**, 10, October 2018, pp. 4951-4977.

DOI: 10.3934/dcds.2018216. [arXiv:1709.00515\[math.PR\]](#).

- Haoyi Xiong, Wei Cheng, Wenqing Hu, Jiang Bian, Zhishan Guo, DBSDA: Lowering the Error Bound of Linear Discriminant Analysis via De-Biasing. *IEEE Transactions on Neural Networks and Learning Systems*, **30**, 3, pp. 707-717, March 2019.

DOI: 10.1109/TNNLS.2018.2846783.

- Wenqing Hu, Vladimír Šverák, Dynamics of geodesic flows with random forcing on Lie groups with left-invariant metrics. *Journal of Nonlinear Science*. **28**, 6, pp. 2249-2274, December 2018.

DOI: 10.1007/s00332-018-9446-1. [arXiv:1510.05279\[math.AP\]](#).

- Wenqing Hu, Itô's formula, the stochastic exponential and change of measure on general time scales. *Abstract and Applied Analysis*, Vol. 2017, Article ID 9140138, 2017.

DOI: 10.1155/2017/9140138. [arXiv:1609.05967\[math.PR\]](#).

- Wenqing Hu, Konstantinos Spiliopoulos, Hypocoelliptic multiscale Langevin diffusions: Large deviations, invariant measures and small mass asymptotics. *Electronic Journal of Probability*, 2017, Vol. 22, article no. 55, pp. 1-38.

DOI: 10.1214/17-EJP72. [arXiv:1506.06181\[math.PR\]](#).

- Tarek Elgindi, Wenqing Hu, Vladimír Šverák, On 2-dimensional incompressible Euler equations with partial damping. *Communications in Mathematical Physics*, **355**, 1, October 2017, pp. 145-159.

DOI: 10.1007/s00220-017-2877-y. [arXiv:1511.02530\[math.AP\]](#).

- Wenqing Hu, Lucas Tcheuko, Random perturbations of dynamical systems with reflecting boundary and corresponding PDE with a small parameter. *Asymptotic Analysis*, **87**, No. 1-2, 2014, pp.

43–56.

DOI: 10.3233/ASY-131197. [arXiv:1203.5092\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, Wave front propagation for a reaction-diffusion equation in narrow random channels. *Nonlinearity*, **26**, 8, 2013, pp. 2333–2356.

DOI: 10.1088/0951-7715/26/8/2333. [arXiv:1303.6943\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, On second order elliptic equations with a small parameter. *Communications in Partial Differential Equations*, **38**, 10, 2013, pp. 1712–1736.

DOI: 10.1080/03605302.2013.812658. [arXiv:1203.5096\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, On diffusion in narrow random channels. *Journal of Statistical Physics*, **152**, 2013, pp. 136–158.

DOI: 10.1007/s10955-013-0763-3. [arXiv:1210.5226\[math.PR\]](#).

- Wenqing Hu, On metastability in nearly-elastic systems. *Asymptotic Analysis*, **79**, 1–2, 2012, pp. 65–86.

DOI: 10.3233/ASY-2011-1090. [arXiv:1202.0577\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, Alexander Wentzell, Small mass asymptotic for the motion with vanishing friction. *Stochastic Processes and their Applications*, **123** (2013), pp. 45–75.

DOI: 10.1016/j.spa.2012.08.013. [arXiv:1201.1242\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, Smoluchowski–Kramers approximation in the case of variable friction. *Journal of Mathematical Sciences*, **79**, 1, November 2011, translated from *Problems in Mathematical Analysis*, **61**, October 2011 (in Russian).

DOI: 10.1007/s10958-011-0589-y. [arXiv:1203.0603\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, On perturbations of generalized Landau–Lifshitz dynamics. *Journal of Statistical Physics*, **144**, 2011, pp. 978–1008.

DOI: 10.1007/s10955-011-0289-5. [arXiv:1203.0602\[math.PR\]](#).

- Mark Freidlin, Wenqing Hu, On stochasticity in nearly-elastic systems. *Stochastics and Dynamics*, **12**, 3, 2012.

DOI: 10.1142/S0219493711500201. [arXiv:1203.5468\[math.PR\]](#).

CONFERENCE PUBLICATIONS

- Jiali Zhang, Thomas S. White, Haoliang Zhang, Wenqing Hu, Donald C. Wunsch, Jian Liu, MS-YOLO: Infrared Object Detection for Edge Deployment via MobileNetV4 and SlideLoss. *International Joint Conference on Neural Networks (IJCNN) 2025, Rome, Italy, June 30 - July 5, 2025*.

- Wenqing Hu, Tiefeng Jiang, Birendra Kathariya, Vikram Abrol, Jiali Zhang, Zhu Li, Subspace Interpolation and Indexing on Stiefel and Grassmann Manifolds as a Lightweight Inference Engine. *IEEE Big Data 2023 (2023 IEEE International Conference on Big Data), Sorrento, Italy, December 15-18, 2023*. Acceptance Rate: 92/526=17.5%.

• Jingfeng Wu, Wenqing Hu, Haoyi Xiong, Jun Huan, Vladimir Braverman, Zhanxing Zhu, On the Noisy Gradient Descent that Generalizes as SGD. *ICML 2020 (37th International Conference on Machine Learning)*, virtual conference due to COVID-19, July 12-18, 2020.

• Huizhuo Yuan, Xiangru Lian, Li, Chris Junchi Li, Ji Liu, Wenqing Hu, Efficient Smooth Non-Convex Stochastic Compositional Optimization via Stochastic Recursive Gradient Descent. *NeurIPS 2019 (Thirty-third Conference on Neural Information Processing Systems)*, Vancouver, Canada, December 8-14, 2019.

• Wenqing Hu, Chris Junchi Li, Xiang Zhou, On the Global Convergence of Continuous-Time Stochastic Heavy-Ball Method for Nonconvex Optimization. *IEEE Big Data 2019 (2019 IEEE International Conference on Big Data)*, Los Angeles, California, USA, December 9-12, 2019.

DOI: 10.1109/BigData47090.2019.9005621

• Md Monirul Islam, Zeyi Sun, Wenqing Hu, Cihan H Dagli, A Framework of Integrating Manufacturing Plants in Smart Grid Operation: Manufacturing Flexible Load Identification. *ICPR 2019 (the 25th International Conference on Production Research)*, Chicago, Illinois, USA, August 10-14, 2019.

DOI: <https://doi.org/10.1016/j.promfg.2020.01.346>

• Wenqing Hu, Zeyi Sun, Yunchao Zhang, Yu Li, Joint Manufacturing and Onsite Microgrid System Control Using Markov Decision Process and Neural Network Integrated Reinforcement Learning. *ICPR 2019 (the 25th International Conference on Production Research)*, Chicago, Illinois, USA, August 10-14, 2019. In press at *Procedia Manufacturing*.

DOI: <https://doi.org/10.1016/j.promfg.2020.01.345>

• Haoyi Xiong, Wei Cheng, Wenqing Hu, Jiang Bian, Yanjie Fu, Zhishan Guo, De-Biasing Covariance Regularized Fisher's Linear Discriminant Analysis with Faster Asymptotic Rate. *IJCAI-ECAI-18 (The 27th International Joint Conference on Artificial Intelligence and the 23rd European Conference on Artificial Intelligence)*, Stockholm, Sweden, July 13-19, 2018.

DOI: <https://doi.org/10.24963/ijcai.2018/401>

• Jiang Bian, Haoyi Xiong, Wei Cheng, Yanjie Fu, Wenqing Hu, Zhishan Guo, Multi-Party Sparse Discriminant Learning. *ICDM 2017 (2017 IEEE International Conference on Data Mining)*, New Orleans, Louisiana, USA, November 8-21, 2017.

DOI: 10.1109/ICDM.2017.86.

• Haoyi Xiong, Wei Cheng, Jiang Bian, Wenqing Hu, Zhishan Guo, AWDA: An Adaptive Wishart Discriminant Analysis. *ICDM 2017 (2017 IEEE International Conference on Data Mining)*, New Orleans, Louisiana, USA, November 8-21, 2017.

DOI: 10.1109/ICDM.2017.62.

• Pengfei Wang, Guannan Liu, Yanjie Fu, Wenqing Hu, Charu Aggarwal, Human Mobility Synchronization and Trip Purpose Detection with Mixture of Hawkes Processes. *KDD 2017 (Knowledge, Discovery and Data mining)*, Halifax, Nova Scotia-Canada, August 13-17, 2017. Accepted paper ID=fp1019.

DOI: 10.1145/3097983.3098067.

UNPUBLISHED
MANUSCRIPT

- Huizhuo Yuan, Wenqing Hu, Stochastic Recursive Momentum Method for Non-Convex Compositional Optimization. Preprint.

arXiv:2006.01688[math.OC]

- Wenqing Hu, Zhanxing Zhu, Haoyi Xiong, Jun Huan, Quasi-potential as an implicit regularizer for the loss function in the stochastic gradient descent. Preprint. Submitted.

arXiv:1901.06054[cs.LG].

AWARDS AND
GRANTS

- Simons Foundation Collaboration Grants for Mathematicians, 2020-2025. Amount=\$42000.
- Miner Alumni Association's Class of '42 Excellence in Teaching Award. June 20, 2018. Amount=\$2000.
- NSF sponsored AMS Travel Support to the International Congress of Mathematicians (ICM) in Rio de Janeiro, Brazil, in August of 2018. Amount=\$3300.
- University of Missouri Research Board. June 1, 2017–May 31, 2018. Topic: Multiscale Stochastic Differential Equations. Amount=\$5000.
- AMS–Simons Travel Grant. July 1, 2015–June 30, 2017. Amount=\$4000.
- Patrick and Marguerite Sung Fellowship in Mathematics. College of Computer, Mathematical and Natural Sciences, University of Maryland, College Park, Spring 2012.
- Block–Grant Graduate Student Fellowship. Department of Mathematics, University of Maryland, College Park, Fall 2008–Spring 2010.

INVITED TALKS/
PRESENTATIONS/
LECTURES

- The 2025 China-UK-US International Symposium on Innovative Nursing Research & Postgraduate Supervisor Training. Soochow University Medical School. Soochow, Jiangsu Province, China. 10/14/2025.

Talk Title: Subspace Interpolation and Indexing on Stiefel and Grassmann Manifolds as a Lightweight Inference Engine.

- Conference “Modern Topics in Probability” in honor of Professor Mark Freidlin’s 85th birthday. Department of Mathematics, University of Maryland. College Park, Maryland, USA. 10/21/2022.

Talk Title: On the Posterior Distribution of a Random Process Conditioned on Empirical Frequencies of a Finite Path: the i.i.d and finite Markov chain case.

- Probability and Statistics Seminar. (online via zoom). Department of Mathematics, University of Kansas. Lawrence, Kansas, USA. 05/04/2022.

Talk Title: On the Posterior Distribution of a Random Process Conditioned on Empirical Frequencies of a Finite Path: the i.i.d and finite Markov chain case.

- Probability Seminar. (online via zoom). Department of Mathematics, University of Washington at Seattle. Seattle, Washington, USA. 04/11/2022.

Talk Title: On the Posterior Distribution of a Random Process Conditioned on Empirical Frequencies of a Finite Path: the i.i.d and finite Markov chain case.

- AMS Sectional Meeting. (online via zoom).

Recent Developments of Variational Methods in Deterministic and Stochastic Systems for the AMS Spring Central Sectional Meeting (Mar 26-27, 2022).

Indiana, USA.

03/26/2022.

Talk Title: On the Posterior Distribution of a Random Process Conditioned on Empirical Frequencies of a Finite Path: the i.i.d and finite Markov chain case.

- Probability Seminar.

Beijing International Center of Mathematical Research (BICMR), Peking University.

Beijing, China.

10/21/2021.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

- Probability Seminar.

School of Mathematical Sciences, Beijing Normal University.

Beijing, China.

10/18/2021.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

- Probability Seminar. (online via zoom).

Department of Mathematics, University of Illinois–Urbana Champaign.

Urbana Champaign, Illinois, USA.

04/27/2021.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

- Stochastics+Data+Computing Seminar (online via zoom).

College of Computing, Department of Applied Mathematics, Illinois Institute of Technology.

Chicago, Illinois, USA.

12/18/2020.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

- Differential Equation Seminar (online due to COVID-19).

Department of Mathematics, University of Missouri (Mizzou).

Columbia, Missouri, USA.

10/08/2020.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

- Duke–Kunshan University.

Kunshan, Soochow, China.

07/14/2020–08/13/2020.

Zu–Chongzhi Lectures at Duke–Kunshan University: Introductory Lectures on Machine Learning, Nonlinear Optimization and Reinforcement Learning. Lectures delivered online.

- Analysis Seminar,

Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).

Rolla, Missouri, USA.

10/21/2019.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

• Department Colloquia,
Department of Statistics, University of Missouri.
Columbia, Missouri, USA. 10/14/2019.

Talk Title: Some Probabilistic Understandings of the Effects of Noise in the Stochastic Gradient Descent.

• Probability and Related Fields Seminar,
Department of Mathematics, Indiana University Bloomington.
Bloomington, Indiana, USA. 10/07/2019.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

• Probability Seminar,
Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA. 08/30/2019.

Talk Title: Wave propagation for reaction-diffusion equations on infinite random trees.

• Seminar at the School of Data Science,
City University of Hong Kong.
Hong Kong, China. 07/03/2019.

Talk Title: Some Probabilistic Understandings of the Effects of Noise in the Stochastic Gradient Descent.

• Seminar on Pure Mathematics,
Department of Mathematics, The Hong Kong University of Science and Technology.
Hong Kong, China. 07/02/2019.

Talk Title: On 2d incompressible Euler equations with partial damping and some related model problems.

• Seminar of Department of Mathematics,
Department of Mathematics, Universidad de Macau (University of Macau).
Macau, China. 06/27/2019.

Talk Title: Some Probabilistic Understandings of the Effects of Noise in the Stochastic Gradient Descent.

• Invited Lecture Series at the Faculty of Science and Technology.
Department of Mathematics, Universidad de Macau (University of Macau).
Macau, China. 06/24/2019 – 06/26/2019.

Invited Lectures: Lectures on Nonlinear Optimization in Machine Learning. 3 lectures, 2 hours each. Lecture notes are available on my personal webpage.

• Applied Computational Intelligence Laboratory.
Department of Electrical and Computer Engineering, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 04/30/2019.

Talk Title: Stochastic Approximations, Diffusion Limit and Small Random Perturbations of Dynamical Systems – a probabilistic approach to machine learning.

- Analysis Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 04/15/2019.

Talk Title: Stochastic Approximations, Diffusion Limit and Small Random Perturbations of Dynamical Systems – a probabilistic approach to machine learning.

- Analysis Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 03/18/2019.

Talk Title: On 2d incompressible Euler equations with partial damping and some related model problems.

- Department Colloquium.
Department of Mathematics, Louisiana State University.
Baton Rouge, Louisiana, USA. 01/25/2019.

Talk Title: Stochastic Approximations, Diffusion Limit and Small Random Perturbations of Dynamical Systems – a probabilistic approach to machine learning.

- Duke–Kunshan University.
Kunshan, Soochow, China. 01/08/2019.

Talk Title: Stochastic Approximations, Diffusion Limit and Small Random Perturbations of Dynamical Systems.

- PDE Seminar.
Department of Mathematics, Vanderbilt University.
Nashville, Tennessee, USA. 11/16/2018.

Talk Title: On 2d incompressible Euler equations with partial damping and some related model problems.

- The 4th Annual Meeting of SIAM Central States Session.
University of Oklahoma.
Norman, Oklahoma, USA. 10/06/2018.

Talk Title: On the long–time behavior of a perturbed conservative system with degeneracy.

- 2018 Anhui Normal University Summer Lecture Series.
School of Mathematics and Statistics, Anhui Normal University.
Wuhu, Anhui, China. 06/27/2018 – 07/08/2018.

Invited Lectures: Lectures on Nonlinear Optimization in Machine Learning. 4 lectures, 2 hours each. Lecture notes are available on my personal webpage.

- PDE Seminar.
School of Mathematics and Statistics, Huazhong University of Science and Technology.
Wuhan, Hubei, China. 06/23/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in opti-

mization.

- Stochastic Analysis Seminar.

College of Mathematics, Sichuan University.
Chengdu, Sichuan, China.

06/22/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Information and Software Engineering Academic Forum.

Department of Computer Science, University of Electronic Science and Technology of China.
Chengdu, Sichuan, China.

06/21/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Randomized Numerical Linear Algebra Seminar.

College of Mathematics and Statistics, Chongqing University.
Chongqing, China.

06/19/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Probability Seminar.

School of Mathematical Sciences, Beijing Normal University.
Beijing, China.

06/12/2018.

Talk Title: Hypocoelliptic multiscale Langevin diffusions and slow-fast stochastic reaction diffusion equations.

- Probability Seminar.

School of Mathematical Sciences, Peking University.
Beijing, China.

06/11/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Invited Lecture “Scientific Frontiers of the 21st Century”,

Beijing Institute of Technology, Beijing, China.

06/08/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Baidu Big Data Lab in Beijing, Baidu, Inc. Beijing, China.

06/07/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

- Stochastic Analysis Seminar.

School of Mathematical Sciences, Peking University.
Beijing, China.

06/05/2018.

Talk Title: On 2-d incompressible Euler equations with partial damping.

• Shanghai Center for Quantitative Life Sciences. Shanghai University.
Shanghai, China. 05/27/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

• Probability Seminar.
Department of Mathematics, University of Wisconsin–Madison.
Madison, Wisconsin, USA. 03/15/2018.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

• Analysis Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 03/12/2017.

Talk Title: A random perturbation approach to some stochastic approximation algorithms in optimization.

• Analysis Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 12/04/2017.

Talk Title: Large deviations and averaging for systems of slow–fast stochastic reaction–diffusion equations.

• Differential Equation Seminar.
Department of Mathematics, University of Missouri (Mizzou).
Columbia, Missouri, USA. 11/02/2017.

Talk Title: On the fast convergence of random perturbations of the gradient flow.

• Analysis Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 09/25/2017.

Talk Title: On the fast convergence of random perturbations of the gradient flow.

• Probability Seminar.
Department of Mathematics, University of Illinois–Urbana Champaign.
Urbana Champaign, Illinois, USA. 09/12/2017.

Talk Title: On the fast convergence of random perturbations of the gradient flow.

• Department of Computer Science, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. Spring 2017.

Invited Lectures: Lectures on the Nature of Statistical Learning Theory. 5 lectures, 1 hour each. Lecture notes are available on my personal webpage.

- Mathematical Association of America–Missouri S&T Chapter.
Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 12/01/2016.

Talk title: From Brownian motion to stochastic calculus, and beyond.

- Kappa Mu Epsilon National Mathematics Honor Society S&T Chapter.
Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 10/12/2016.

Talk title: From Brownian motion to stochastic calculus, and beyond.

- Time Scales Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 10/05/2016.

Talk title: Itô's formula, the stochastic exponential and change of measure on general time scales.

- The 2nd Annual Meeting of SIAM Central States Section.
University of Arkansas at Little Rock.
Little Rock, Arkansas, USA. 10/01/2016.

Talk title: On 2–d incompressible Euler equations with partial damping.

- Computational and Applied Mathematical Sciences Seminar.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 09/27/2016.

Talk title: Hypocoelliptic multiscale Langevin diffusions: Large deviations, invariant measures and small mass asymptotics.

- Stochastics Seminar.
School of Mathematics, Georgia Institute of Technology.
Atlanta, Georgia, USA. 04/07/2016.

Talk title: Dynamics of geodesic flows with random forcing on Lie groups with left–invariant metrics.

- Department Colloquium.
Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla).
Rolla, Missouri, USA. 02/19/2016.

Talk title: Stochastically Perturbed Geodesic Flows on Lie Groups.

- Probability Seminar.
School of Mathematics, University of Minnesota, Twin Cities.
Minneapolis, Minnesota, USA. 01/29/2016.

Talk title: Dynamics of geodesic flows with random forcing on Lie groups with left–invariant metrics.

- 2015 Peking University Youth Probability Forum.
Peking University, Beijing, China. 07/09/2015 – 07/17/2015.

Invited Lectures: Lectures on Stochastic Fluid Mechanics. 5 lectures, 2 hours each. Lecture notes are available on my personal webpage.

- 2015 Peking University Youth Probability Forum.
Peking University, Beijing, China. 07/08/2015.

Talk title: Random motion along co-adjoint orbits.

- AMS 2015 Spring Sectional Meeting.
Georgetown University. Special Session on Asymptotic Problems for Stochastic Processes and PDEs.
Washington D.C., USA. 03/08/2015.

Talk title: Random motion along co-adjoint orbits.

- Cincinnati Symposium on Probability Theory and Applications.
Department of Mathematical Sciences, University of Cincinnati.
Cincinnati, Ohio, USA. 09/20/2014.

Poster Presentation: Second order elliptic equations with a small parameter.

- Workshop “Recent Advances in PDEs and Fluids”.
Department of Mathematics, Stanford University.
Palo Alto, California, USA. 08/17/2013.

Talk title: On diffusion and wave front propagation in narrow random channels.

- The 9th Cornell Probability Summer School.
Department of Mathematics, Cornell University.
Ithaca, New York, USA. 07/25/2013.

Talk title: On diffusion and wave front propagation in narrow random channels.

- Conference on Asymptotic Problems in Stochastic Processes and PDEs in honor of Professor Freidlin’s birthday.
Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA. 05/21/2013.

Talk title: On diffusion and wave front propagation in narrow random channels.

- PDE Seminar.
School of Mathematics, University of Minnesota, Twin Cities.
Minneapolis, Minnesota, USA. 12/12/2012.

Talk title: Second order elliptic equations with a small parameter.

- Applied PDE Research Interaction Team.
Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA. 12/03/2012.

Talk title: Second order elliptic equations with a small parameter.

- The 8th Cornell Probability Summer School.
Department of Mathematics, Cornell University.
Ithaca, New York, USA. 07/26/2012.

Talk title: Small mass asymptotic for the motion with variable and vanishing friction.

- Organizing the Student Probability Seminar.

Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA.

Summer 2012.

Giving series of talks on stochastic partial differential equations.

- Organizing the Student Probability/Combinatorics Seminar.

Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA.

Fall 2011, Spring 2012.

Giving series of talks: Basic theory; The second moment method (I), (II), (III); The local lemma (I), (II), (III); Martingales and tight concentration (I), (II), (III).

- Student Analysis/PDE/Probability Seminar.

Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA.

12/08/2010.

Talk title: An introduction to Schramm–Löwner evolutions.

- Student Analysis/PDE/Probability Seminar.

Department of Mathematics, University of Maryland, College Park.
College Park, Maryland, USA.

03/24/2010.

Talk title: Probabilistic approach to some PDE problems.

ACADEMIC
SERVICE/
MEMBERSHIP

- I serve as a reviewer of *Mathematical Reviews* (MathSciNet, Reviewer Number: 092509).

• I served as referee for the following journals: *Nonlinearity*, *Transactions of the American Mathematical Society*, *Discrete and Continuous Dynamical Systems–Series A* (3 times), *Journal of Nonlinear Science*, *Stochastic Processes and their Applications* (3 times), *Journal of Differential Equations* (4 times), *Punjab University Journal of Mathematics*, *Journal of Theoretical Probability* (3 times), *Journal of Mathematics and Statistics*, *Acta Mathematica Scientia*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Image Processing* (3 times), *Asymptotic Analysis*, *SIAM Journal on Mathematical Analysis*, *Annals of Applied Probability* (4 times), *Stochastics and Partial Differential Equations: Analysis and Computations*, *Potential Analysis*, *Applied Probability Journals*, *Probability Theory and Related Fields*.

• I served as reviewer for the following conferences: *Neural Information and Processing Systems (2019/2022-2024)*, *International Conference on Learning Representation (2020/2021/2024)*, *International Conference on Machine Learning (2022-2026)*, *The AAAI conference on Artificial Intelligence (2019/2025/2026)*, *IEEE Big Data Conference (2019/2023/2024)*, *International Conference on Image Processing (2022)*, *IEEE International Conference of Data Mining (ICDM, 2025)*.

• I served as the Program Committee Vice Chair for *IEEE Big Data Conference in 2019* in charge of the Big Data Science and Foundations area.

• I served as the Program Committee Member for *AAAI 2021 (35th AAAI Conference on Artificial Intelligence)*, *IEEE-BigData 2023*, *AAAI 2025/2026*.

• I served as an External Reviewer for *Canada Research Chair* position in 01/2023.

TEACHING

- Courses Taught at Missouri University of Science and Technology (formerly University of Missouri, Rolla): Calculus III, Linear Algebra, Elementary Differential Equations, Partial Differential Equations, Introduction to Real Analysis, Introduction to Complex Variables, Nonlinear Optimization in Machine Learning, Computational Optimization.
- Courses Taught at University of Minnesota, Twin Cities: Applied Linear Algebra, Introduction to Stochastic Processes, Basic Theory of Probability and Statistics, Advanced Calculus I.
- Courses involved as either Teaching Assistant or Grader or Tutor at University of Maryland, College Park: Calculus I-III, Real Analysis, Linear Algebra, Linear Algebra for Scientists and Engineers, Advanced Calculus I-II, Probability, PDE for Scientists and Engineers, Complex Analysis for Scientists and Engineers, Introduction to Numerical Analysis, Abstract Algebra.

GRADUATE ADVISING

I am advising the following students at Missouri University of Science and Technology (formerly University of Missouri, Rolla):

- Huang, Qitong, Master's degree, 2026–now, currently working with me on generative models via stochastic differential equations.
- Zhang, Jiali. PhD, 2022–now, currently working with me on image processing, reinforcement learning and advanced manufacturing.

I have advised the following students at Missouri University of Science and Technology (formerly University of Missouri, Rolla):

- Gozebina, Erich., Master's degree. Thesis: Transformer-based language models for Bitcoin market prediction and interpretation. Graduation Date: May 2026.
- O'Conner, Paul K., Master's degree. Thesis: Cryptographic Algorithms, Cryptocurrencies, and a predictive model of bitcoin value by PLS regression. Graduation Date: May 2024.
- Steimeister, Louis K., PhD, 2019–2020, has been working with me on reinforcement learning and advanced manufacturing. Now at TU Dortmund in Germany for a second PhD.
- Vandergriffe, Austin., Master's degree. Graduation Date: August 2021.

I also served as a member of the PhD/Master's thesis defense committee for various doctoral students at Missouri University of Science and Technology (formerly University of Missouri, Rolla).

UNDERGRADUATE ADVISING

I advised the following students on their senior projects – typically an expository paper on a subject chosen by the student.

At Missouri University of Science and Technology (formerly University of Missouri, Rolla):

- Spring 2026. MATH 5325 Partial Differential Equation–Honors Program. Term Paper Title: Understanding the Geometry and Mechanics Behind 2D Fluid Motion. Student: Baylor Dickinson.
- Spring 2018. MATH 3304H–1L. Elementary Differential Equations–Honors Program. Term Paper Title: Lotka–Volterra models of Predator–Prey Relationships. Student: Tyler Blaszkak, ID=12539004.

At University of Minnesota, Twin Cities:

- Spring 2016. Senior project for independent study (MATH 4997W). Adviser for Wu, Zhouman and Li, Weiqian. Topic: VaR (Value at Risk) estimation and extreme value theory.
- Spring 2016. Senior project for independent study (MATH 4995). Adviser for Yang, Wenjing. Topic: An overview of linear model selection methods.
- Fall 2015. Senior project for independent study (MATH 4997W). Adviser for Xu, Yitong and Sun, Xi. Topic: A review of the Capital Asset Pricing Model.
- Fall 2013. Senior project for independent study (MATH 4997W). Adviser for Lin, Htet Naing and Htet, Maung Soe. Topic: A review of the theory of Markov chains.

I supervised the following undergraduate teams participating the Mathematical Contest in Modeling (MCM).

At University of Minnesota, Twin Cities:

- MCM 2016. Team number: 42730. Team members: Hou, Yucheng; Zhu, Xiaoyi; Song, Qixin. Topic: An ODE model for the temperature dynamics of bathtub system with a person sitting in – the evolution of temperature for different configurations and feedback control mechanism. Result: Honorable Mention.
- MCM 2015. Team number: 42027. Team members: Liu, Jiaoyue; Wang, Xiaoqing; Teng, Da. Topic: The model based on SEIR for Ebola.

DEPARTMENT/ UNIVERSITY SERVICE

- Department Committee, Department of Mathematics and Statistics, Missouri University of Science and Technology (formerly University of Missouri, Rolla): Department Policy Committee, 09/2017–12/2019. Department Website Committee, 08/2019–now. Department Graduate Admissions Committee, 01/2021–08/2022. Department Data Sciences Program Committee, 09/2021–now. Department NTT Promotion Committee, 10/2022–now. Department Colloquium Chair, 02/2023–now.
- Campus Faculty Senate, Missouri University of Science and Technology (formerly University of Missouri, Rolla), 09/2022–now.
- Associate Investigator, Intelligent Systems Center (ISC), Missouri S&T, 07/2020–now.

INDUSTRIAL EMPLOYMENT

- Consulting work on cryptographic zero-knowledge proofs of Ethereum Virtual Machines (EVM) at Scroll Tech during 2023-2024: Scroll is the leading zero-knowledge rollup. Scaling Ethereum for good. See <https://scroll.io/>

PROFESSIONAL SKILLS/ CERTIFICATIONS

- Programming Language: Python, Rust, C/C++, MATLAB, Java, mySQL, R.
- Operating System: Windows 10, Linux (Ubuntu 22.04), xv6.
- Professional Certificate: Society of Actuaries (SOA) Exams (P=Probability, Taken September 2015, Status=Pass; FM=Financial Mathematics, Taken June 2016, Status=Pass).
- Language: English (Proficient), Chinese (Native).

Last updated: September 2, 2026.